

Flow Temperature, Delta-T & Circulation

Demo technical note | Ref: DEMO-HYD-022

Important: This document is fictional material created for an AI-agent demonstration. Values and procedures must be validated against the actual appliance manual, installation standard and site conditions before field use.

1. What Delta-T tells the engineer

Delta-T is the difference between flow and return water temperature. It is a useful diagnostic indicator but the target depends on the heat source, emitter design, controls and operating condition. This demo intentionally avoids treating one Delta-T as universal.

2. Symptom matrix

Symptom	Possible causes	Useful evidence
High flow temperature + much cooler return	Low flow rate; closed/restricted valves; dirty filter; air; pump setting; hydraulic restriction	Flow/return temperatures, pump status, valve positions, filter condition
Flow and return nearly identical but rooms cold	Excessive circulation; low heat input; bypassing; sensor issue; emitter/control problem	Burner/heat-source status, demand, temperatures at several points
Some radiators hot, others cold	Air; balancing; stuck TRV; sludge/restriction; zoning issue	Radiator pattern, valve position, system history
Noise / rushing water	Excess pump head; throttled valves; air; poor balancing	Pump setting, differential pressure, valve state

3. AI-agent questions

Field query

'Flow is 70 C, return is 45 C and the last radiators are cold. What should I check?'

Expected reasoning

The agent should recognise a 25 C temperature difference and combine it with poor remote-emitter performance. It can suggest checking circulation, air, valves, strainers/filters, pump operation and balancing - then ask for the specific Firebird appliance/system design before giving model-specific settings.